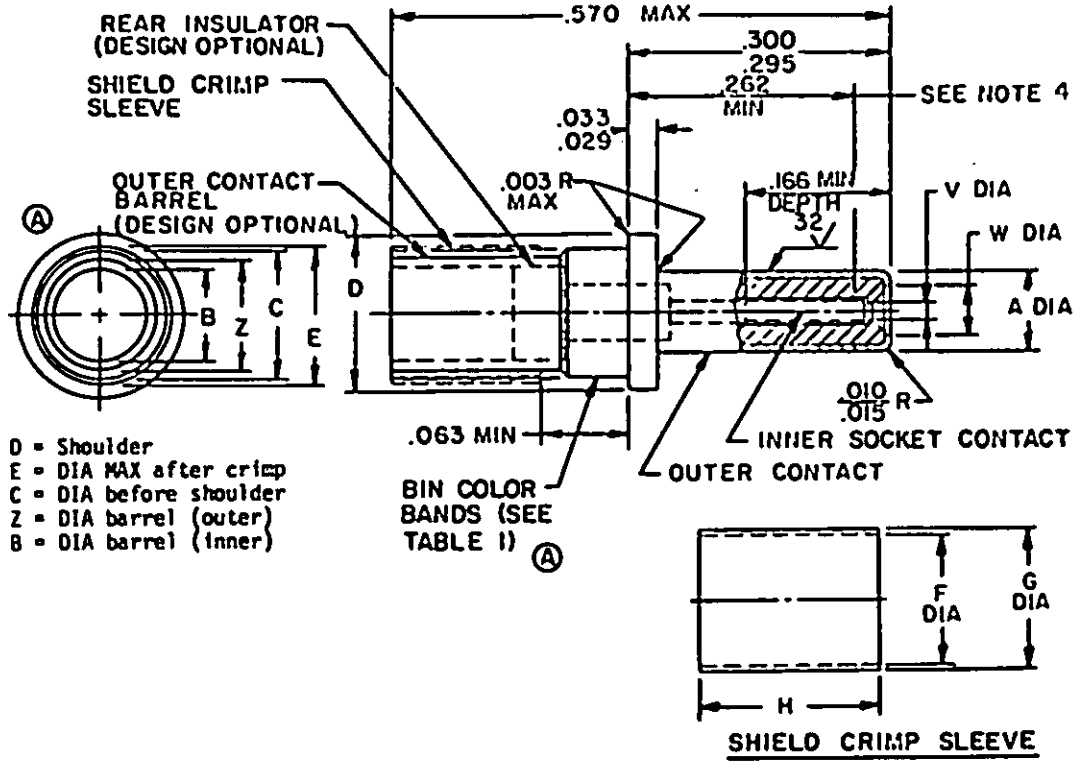
MILITARY SPECIFICATION SHEET

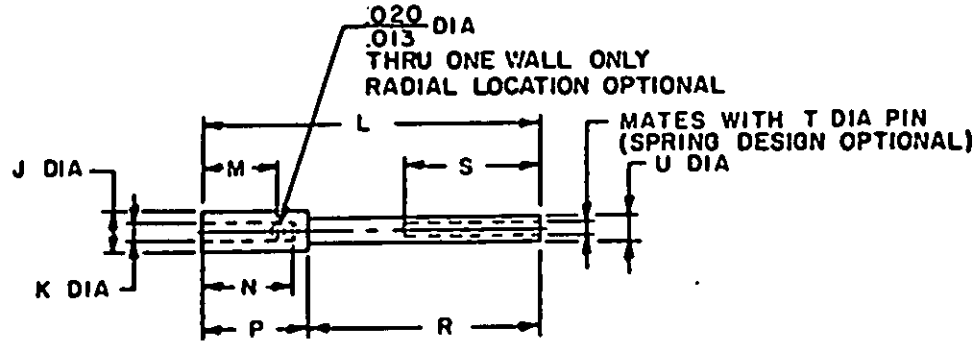
CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, SHIELDED, SIZE 16 (FOR MIL-C-38999 SERIES I, II, III, IV AND MIL-C-24308 CONNECTORS)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the contacts described herein shall consist of this document and the latest issue of Specification MIL-C-39029.



SHIELD CRIMP SLEEVE



INNER SOCKET CONTACT

Ⓐ Denotes Changes

FIGURE 1. Pin contacts.

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.

BIN code	A Dia.	B Dia. Min.	C Dia.	D Dia.	E Dia. Max.	F Dia. Min.	G Dia. Max.	H	J Dia. Max.	K Dia. Min.	L REF
424	.0635 .0615	.0670	.103 .101	.130 .127	.108	.105	.120	.105 .095	.046	.0210	.392
425	↓	.0575	↓	↓	↓	.094	↓	↓	.046	.0210	↓
426	↓	.0670	↓	↓	↓	.105	↓	↓	.052	.0355	↓
427	.0635 .0615	.0575	.103 .101	.130 .127	.108	.094	.120	.105 .095	.046	.0270	.392

BIN code	M	N Min.	P	R	S Min.	T	U	V Dia.	W Dia.	Z Dia. Max.
424	.094 .087	.103	.125 .119	.271 .269	.156	.0155 .0145	.030 .028	.022 .018	.041 .038	.085
425	↓	↓	↓	↓	↓	↓	↓	↓	↓	.076
426	↓	↓	↓	↓	↓	↓	↓	↓	↓	.085
427	.094 .087	.103	.125 .119	.271 .269	.156	.0155 .0145	.030 .028	.022 .018	.041 .038	.076

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.027	.69	.046	1.17	.085	2.16	.119	3.02	.166	4.22
.010	.25	.028	.71	.052	1.32	.087	2.21	.120	3.05	.262	6.65
.013	.33	.029	.74	.0575	1.460	.094	2.39	.125	3.18	.269	6.83
.015	.38	.030	.76	.0615	1.562	.095	2.41	.127	3.23	.271	6.88
.018	.46	.033	.84	.063	1.60	.101	2.57	.130	3.30	.295	7.49
.020	.51	.0355	.902	.0635	1.613	.103	2.62	.145	3.68	.300	7.62
.021	.53	.038	.97	.067	1.70	.105	2.67	.155	3.94	.392	9.96
.022	.56	.041	1.04	.076	1.93	.108	2.74	.156	3.96	.570	14.48

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Dimensions shown apply after plating.
4. Point at which a square ended pin of the same basic diameter as the mating contact first engages the inner contact spring. Provision for clearance hole shall be provided.
5. Crimp deformation: The maximum diameter over the crimped portion of the shield crimp sleeve shall not exceed E diameter.

FIGURE 1. Pin contacts - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Contacts shall comply with the reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999.

Dimensions, design characteristics, and configuration: See figure 1 and table I.

Mating contacts: MIL-C-39029/77 and MIL-C-39029/78.

Tools: See table II.

TABLE I. Design characteristics.

BIN code	Color bands			Cable accommodated	Contact cavity size	Type	Class
	1st	2nd	3rd				
424	Yellow	Red	Yellow	M17/119-RG174 M17/113-RG316 M17/094-RG179 1/2/ Times AA3248 Teledyne 11299 Thermax 75-738-8CCWXE Tensolite 30888/L707YX-1 Haveg 8100207	16	D	B
425	Yellow	Red	Green	M17/093-RG178			
426	Yellow	Red	Blue	1/ Haveg 61-02051 Revere WH95623 (red shielded)			
427	Yellow	Red	Violet	1/ Haveg 30-00761 Haveg 30-02024 Haveg 30-02033 Tensolite 24713/A955KK1 Tensolite 26723/A955KK1			

1/ Or equivalent.
2/ High tensile strength copper alloy wire.

Ⓐ TABLE II. Tools.

BIN code	Inner contact		Outer contact		Installing tool	Removal tool
	Basic crimping tool	Positioner	Basic crimping tool	Positioner		
424, 425, 426, 427	M22520/2-01	M22520/2-35	M22520/4-01	M22520/4-02	M81969/8-07 or M81969/14-03	M81969/8-08 or M81969/14-03

Ⓐ Cable to contact information: See table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.

Ⓐ TABLE III. Cable to contact information.

BIN code	Cable accommodated	Inner contact tool selector setting no.
424	M17/119-RG174	3
	M17/113-RG316	5
	M17/094-RG179	3
	1/ 2/ Times AA3248	↓
	Teledyne 11299	
	Thermax 75-738-BCCWXE	
	Tensolite 30888/L707YX-1	
	Haveg 8100207	
425	M17/093-RG178	3
426	1/ Haveg 61-02057	4
	Revere WH95623 (red shielded)	4
427	1/ Haveg 30-00761	3
	Haveg 30-02024	4
	Haveg 30-02033	3
	Tensolite 24713/A955KK1	3
	Tensolite 26723/A955KK1	3

1/ Or equivalent.
2/ High tensile strength copper alloy wire.

Contact resistance: See table IV.
Test current:
Inner contact - 1 ampere.
Outer contact - 12 amperes.

Low signal level contact resistance (inner contact only): See table V.

Contact engagement and separation forces (inner socket contact only): The engagement depth shall be as encountered in normal service. The test pins shall be in accordance with MS3197 except the diameters shall be as specified in the following, and surface roughness shall not exceed 3 microinches. Provision for clearance hole shall be provided.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces)		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.0155 $^{+.0002}_{-.0000}$	NA	NA	12.0	14.0	NA
.0145 $^{+.0000}_{-.0002}$	0.5	0.4	NA	NA	NA

Dielectric withstanding voltage (applied between inner and outer contact):
Test voltage:
At sea level - 800 Vac rms.
At 50,000 feet - 250 Vac rms.

Tensile strength (inner and outer contact crimp joint): See table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.

TABLE IV. Contact resistance.

BIN code	Cable accommodated	Maximum voltage drop (millivolts)						Maximum average voltage drop
		25° +3° -0°C		2/ 25° +3° -0°C		200° +3° -0°C		
		Inner contact	Outer contact	Inner contact	Outer contact	Inner contact	Outer contact	
424	M17/119-RG174	55	85	66	102	94 4/	145 4/	NA
	M17/113-RG316	55	75	66	90	94	128	
	M17/094-RG179	120	70	144	84	204	119	
	1/3/ Times AA3248	170	150	204	180	290	255	
	Teledyne 11299	↓	↓	↓	↓	↓	↓	
	Thermax 75-738-BCCWXE							
	Tensolite 30888/L707YX-1							
	Haveg 8100207							
425	M17/093-RG178	120	110	144	132	204	187	
426	1/ Haveg 61-02051	65	100	78	120	111	170	
	Revere WH95623 (red shielded)	65	100	78	120	111	170	
427	1/ Haveg 30-00761	120	110	144	132	204	187	
	Haveg 30-02024	120	110	144	132	204	187	
	Haveg 30-02033	120	110	144	132	204	187	
	Tensolite 24713/A955KK1	55	100	66	120	94	170	
	Tensolite 26723/A955KK1	65	100	78	120	111	170	↓

1/ Or equivalent.
2/ After conditioning.
3/ High tensile strength copper alloy wire.
4/ 85° +3° -0°C.

Ⓐ TABLE V. Low signal level contact resistance (inner contact only) and tensile strength.

BIN code	Cable accommodated	Maximum contact resistance (milliohms)		Tensile load (pounds) (minimum)	
		Initial	After conditioning	Inner contact	Outer contact
424	M17/119-RG174	55	66	15.0	15.0
	M17/113-RG316	55	66	10.0	15.0
	M17/094-RG179	120	144	3.5	15.0
	1/2/ Times AA3248	170	204	7.0	35.0
	Teledyne 11299	↓	↓	↓	↓
	Thermax 75-738-BCCWXE				
	Tensolite 30888/L707YX-1				
	Haveg 8100207				
425	M17/093-RG178	120	144	3.5	10.0
426	1/ Haveg 61-02051	65	78	15.0	15.0
	Revere WH95623 (red shielded)	65	78	15.0	15.0
427	1/ Haveg 30-00761	120	144	4.0	10.0
	Haveg 30-02024	120	144	↓	↓
	Haveg 30-02033	120	144		
	Tensolite 24713/A955KK1	55	66		
	Tensolite 26723/A955KK1	65	78		

1/ Or equivalent.
2/ High tensile strength copper alloy wire.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.

Ⓐ Vibration: Test condition VI, letter J, per method 2005 of MIL-STD-1344. Except directions shall be longitudinal and perpendicular. Time duration shall be 8 hours in each direction.

Ⓐ Shock (specified pulse): Test condition D, per method 2004 of MIL-STD-1344.

Ⓐ Shipping: Figure 1 illustrates final assembly; parts are to be shipped unassembled in a suitable container. Manufacturers recommended assembly instructions shall be shipped with the unit package.

QPL evaluating activity: Defense Electronics Supply Center (DESC-E), Dayton, OH 45444.

Military part number: See table VI.

Ⓐ TABLE VI. Military part number.

BIN code	Military part number
424	M39029/76-16A
425	M39029/76-16B
426	M39029/76-16C
427	M39029/76-16D

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR
Navy - EC, MC, OS
Air Force - 11, 15, 17, 99
DLA - ES

Preparing activity:
Air Force - 85

(Project 5935-3051-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 202610-21T18:42Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

AFLC/DSP
Wright-Patterson AFB OH 45433

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.